Mass and weight – what is which? 15

Mass is defined as the amount of substance in an object. Its base unit is the kilogram (kg).

Weight is a force. It is a force that arises because of gravity. The larger an object, the stronger the gravity force; so huge objects like stars have strong gravity, which causes things to weigh more there than on smaller objects. Weight is measured in newtons (N).

It is common to say 'My **weight** is 50 kg', but this is scientifically incorrect and probably arises from the confusion between mass and weight that many people have. (A person of mass 50 kg actually has a weight of 500 N.) We should actually say 'My **mass** is 50 kg'.

We are going on an imaginary journey through the solar system. As well as food and radiation suits, we are taking a giant triple beam balance for measuring our mass and a bathroom scale for our weight. On earth, the giant triple beam balance shows your mass as 60 kg and the bathroom scale gives your weight as 60 kg (although this is really 600 N!).

1 Let's fly towards the moon. At a certain point between the earth and the moon, the gravitational forces of the earth and moon cancel each other. At that point:

a How strong is the gravitational force on us? _____ (newtons)

b What is your mass? _____ (kilograms)

c What does the bathroom scale show your weight to be? _____ (kilograms)

d What is your weight in scientific measure? _____ (newtons)

2 Now let's land on the moon and take our measuring devices with us onto the moon's surface. The moon's mass is about six times smaller than the earth's mass. This means that the moon's gravity is one-sixth that of the earth's gravity, i.e. the moon's gravity is six times smaller than the earth's.

a What is your mass on the moon, as measured by the giant triple beam balance? _____

b What does the bathroom scale show your weight to be? _____

c What is your scientific weight? _____

d If you can jump 1 metre high on earth, how high could you jump on the moon? _____

3 We're leaving the moon and travelling on to Jupiter, which has a mass about 300 times bigger than the earth.

a About how strong is Jupiter's gravity compared to earth's gravity? _____

b What is your mass on Jupiter, as measured by the giant triple beam balance? _____

c What does the bathroom scale show your weight to be on Jupiter? _____

d If you could lift 30 kg (30 000 g) on earth, what could you lift on Jupiter? _____

4 Onwards to the sun. The sun's surface temperature is about 6 000°C and we would simply burn into a cinder well before arrival; but let's ignore that small detail. The sun's mass is about 300 000 times bigger than the earth's mass.

a About how strong is the sun's gravity force compared to the earth's? _____

b What is your mass on the sun? _____

c What is your weight on the sun? _____

d If you could throw a ball 30 m (= 30 000 mm) into the air on earth, how high could you throw the same ball on the sun? _____

According to Newton's second law of motion, the force on an object depends on its mass and its acceleration, i.e. $F = m \cdot a$. The mass is your body mass (e.g. 60 kg) and the acceleration is the acceleration due to gravity (10 m/s^2 on earth). Therefore the weight force on earth is $F = m \cdot a = 60\ kg \cdot 10\ m/s^2 = 600$ N. Note that on earth, one Newton equals the gravity force on 100 grams; 10 N attract 1 kg.

On Mars, the acceleration due to gravity is only about 1 m/s^2, as Mars has a fairly small mass.

5 **a** What would your weight force be on Mars? _____

b What does the bathroom scale show your weight to be? _____

6 Uranus has a gravity acceleration of about 150 m/s^2, as its mass is much greater than the earth's.

a Calculate your weight force on Uranus. _____

b State your weight as shown on a bathroom scale. _____

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